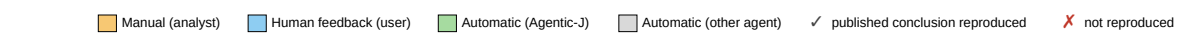
a CD177 neutrophil morphology

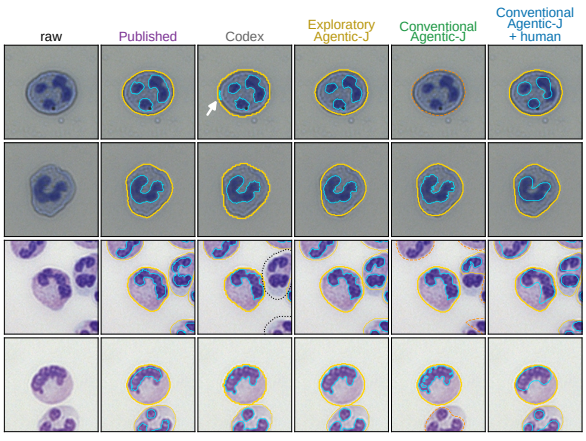
Do CD177+ and CD177− neutrophils differ in cell and nucleus morphology?

	Cell segmentation	Nucleus segmentation	QC & post-processing	Features	Statistics	Time to result	Scientific verdict
Published	Cellpose (default)	ilastik pixel classifier	dilation, crop, size filter	25 hand-picked	cell-level Mann–Whitney	weeks	no meaningful difference
Codex CLI	Cellpose cyto3 at half resolution	erosion/threshold within cell	shape QC	30 features	donor-level Wilcoxon, FDR	13 min	✓ 0/30 features significant
Exploratory Agentic-J	Cellpose cyto3	Otsu threshold within cell	border, size, solidity filter	25 features	donor-level Wilcoxon, FDR	35 min	✗ 5/25 significant, 3/3 runs
Conventional Agentic-J	Cellpose cyto3 on optical density	threshold within cell	size/shape QC	32 features	donor-level permutation, FDR	67 min	✓ 0/32 significant, 3/3 runs
Conventional Agentic-J + human	Cellpose-SAM, chosen by the user	micro-sam fine-tuned on 11 hand-annotated tiles	lobe-to-cell assignment	11 features	image-level Mann–Whitney, FDR	488 min	✓ 0/11 features significant

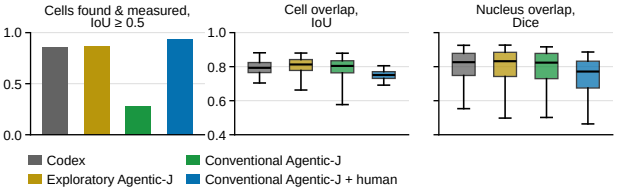


Manual (analyst) · Human feedback (user) · Automatic (Agentic-J) · Automatic (other agent) · ✓ published conclusion reproduced · ✗ not reproduced

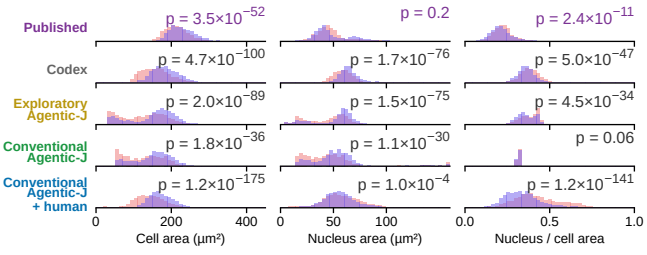
b CD177: segmentations of four example cells



c CD177: agreement with the published masks, 4389 cells



d CD177: per-cell distributions vs published data

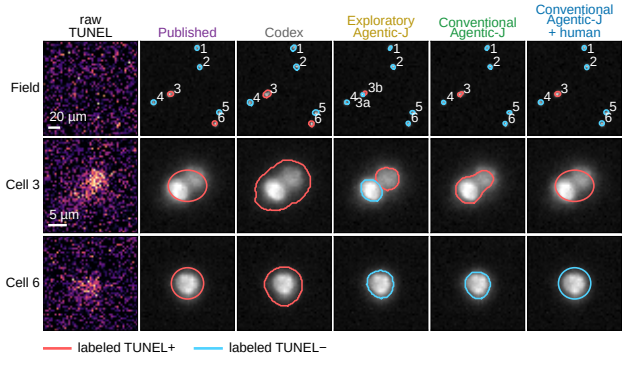


e Leishmania infection

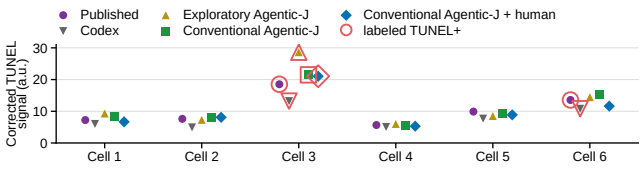
Task 1: call each of 6 host nuclei TUNEL+, meaning apoptotic, or TUNEL−. Task 2: compare the proliferation index of 15 parasites between TUNEL+ and TUNEL− hosts.

	Registration, live → fixed	Host / nucleus segmentation	TUNEL call	Parasite localisation	Measurement & statistics	Time T1 / T2	Task 1 TUNEL+ / TUNEL− calls 6 nuclei, fixed frame	Task 2 proliferation index 15 parasites, last live frame
Published	by hand	hand-drawn ROIs	by inspection	15 ROIs + 3 backgrounds each, by hand	ROI Manager → Excel; Mann–Whitney	not recorded	2/6 nuclei TUNEL+	8 TUNEL+ vs 7 TUNEL−, p = 0.0012
Codex CLI	SIFT + RANSAC, ECC-refined to 1.6 px	T1 DAPI threshold; T2 Cellpose cyto3	largest-gap cutoff	red-channel peaks + water-shed: 10 kept	annulus background; no test	18 min, both tasks	✓ 6/6 calls agree, r = 0.98	✓ 10/15 found, hosts right; p = 0.017
Exploratory Agentic-J	TurboReg rigid on brightfield	T1 StarDist; T2 Cellpose cyto3, 7 hosts	largest-gap cutoff	C2 + C3 spot evidence: 22 kept	annulus background; no test	22 / 69 min	✓ 5/6 calls agree, r = 0.95 only cell 6 differs; cell 3 split in two	✗ 0/15 found, 22 false, no host
Conventional Agentic-J	TurboReg / StackReg rigid	T1 DoG + water-shed; T2 Cellpose cyto3, 6 hosts	2-component GMM	micro-sam vit_b_lm: 105 → 5 kept	annulus background; no test	55 / 85 min	✓ 5/6 calls agree, r = 0.99 only cell 6 differs	✗ 3/15 found, 2 false, no TUNEL+
Conventional Agentic-J + human	lab ROIs given, origin translated	lab ROIs, given	largest-gap cutoff	lab ROIs paired 1:1 with backgrounds	native 16-bit; Mann–Whitney	34 / 26 min	✓ 5/6 calls agree, r = 0.97 only cell 6 differs	✓ 15/15; 14/15 within 0.1; p = 0.0012

f Leishmania task 1: six nuclei, called by each pipeline



g Leishmania task 1: corrected TUNEL signal per nucleus



h Leishmania: counts compared with the published analysis

		Codex CLI	Exploratory Agentic-J	Conventional Agentic-J	Conventional Agentic-J + human
Task 1	TUNEL calls agree	6 / 6	5 / 6	5 / 6	5 / 6
Task 2	Lab parasites found	10 / 15	0 / 15	3 / 15	15 / 15
	Host group correct	10 / 10	–	2 / 3	15 / 15
	Index within 0.1 of lab	4 / 10	–	1 / 3	14 / 15
	False detections	0	22	2	0

i Leishmania task 2: proliferation index by host TUNEL status

